Part 3

- $\hat{y} = m_1 x_1 + m_2 x_2 + b$

Parameters → m_1, m_2, b

$$x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

1) $\hat{y} = x \cdot m + b$

→ $\begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

→ $(1)(-1) + 3(2) = -1 + 6 = 5$

→ $(4)(-1) + 10(2) = -4 + 20 = 16$

$$\hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Error vector → $\hat{y} - y = \begin{matrix} 6 \\ 17 \end{matrix} - \begin{matrix} 5 \\ 6 \end{matrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

2) $MSE = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$

$e = \hat{y}_i - y_i$

$J(m, b) = \frac{1}{n} \cdot \sum e_i^2$

In matrix form.

$J(m, b) = \frac{1}{n} \cdot (e^T \cdot e)$, where $e = x \cdot m + b - y$

$$e = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$J = \frac{1}{2}(1^2 + 11^2) = \frac{1}{2}(1 + 121) = \boxed{61}$$

3) $e_i = m_1 x_{i1} + m_2 x_{i2} + b_i - y_i$

$$J = \frac{1}{n}(e_1^2 + e_2^2)$$

Gradient in respect to m_1

$$\frac{dj}{dm_1} = J = \frac{1}{2}\left[(\hat{y}_1 - y_1)^2 + (\hat{y}_2 - y_2)^2\right]$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$\hat{y}_1 = m_1 x_{11} + m_2 x_{12} + b \qquad x_{11} = 1 \qquad x_{21} = 4$

$\hat{y}_2 = m_1 x_{21} + m_2 x_{22} + b \qquad x_{12} = 3 \qquad x_{22} = 10$

$$J = \frac{1}{2}\left[(m_1 + 3m_2 + b - 5)^2 + (4m_1 + 10m_2 + b)^2\right]$$

$$\frac{dj}{dm_1} = \frac{1}{2}\left[2(m_1 + 3m_2 + b) \times 1 + 2(4m_1 + 10m_2 + b) \times 4\right]$$

$$m_1 + 3m_2 + b + 4(m_1 + 10m_2 + b)$$

$$\frac{dj}{dm_1} = \boxed{e_1 + 4e_2}$$

Derive in respect to m_2

$$J = \frac{1}{2}(m_1 + 3m_2 + b - 5)^2 + (4m_1 + 10m_2 + b - 6)^2$$

$$\frac{dj}{m_2} = \frac{1}{2}\left(2(m_1 + 3m_2 + b - 5) \times 3 + 2(4m_1 + 10m_2 + b - 6) \times 10\right)$$

$$= 3(m_1 + 3m_2 + b - 5) + 10(4m_1 + 10m_2 + b - 6)$$

$$\frac{dJ}{dm_2} = \boxed{3e_1 + 10e_2}$$

Derive in respect to b

$$J = \frac{1}{2}\left[(m_1 + 3m_2 + b - 5)^2 + (4m_1 + 10m_2 + b - 6)^2\right]$$

$$\frac{dj}{db} = \frac{1}{2}\left[2(m_1 + 3m_2 + b - 5) \times 1 + 2(4m_1 + 10m_2 + b - 6)\right]$$

$$= m_1 + 3m_2 + b - 5 + 4m_1 + 10m_2 + b - 6$$

$$\frac{dj}{db} = \boxed{e_1 + e_2}$$

$$\rightarrow \frac{dj}{dm} = \begin{bmatrix} e_1 + 4e_2 \\ 3e_1 + 10e_2 \end{bmatrix} \quad \text{where } e = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 + 4(11) \\ 3 + 10(11) \end{bmatrix} \rightarrow \boxed{\begin{bmatrix} 45 \\ 113 \end{bmatrix}}$$

$\frac{dj}{db} = e_1 + e_2$ where $e = \begin{matrix}1\\11\end{matrix}$

$\Rightarrow 1 + 11 = \boxed{12}$ $\quad \frac{dj}{db} = \begin{bmatrix} e1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

→ m_1, m_2 and b are positive so gradient descent will decrease all three

4) Iteration 0

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad \alpha = 0.01$

$J = 61$

→ new = old − α x gradient

so $m^1 = m^0 - \alpha \times \frac{dj}{dm}$

$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \times \begin{bmatrix} 45 \\ 113 \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.45 \\ 1.13 \end{bmatrix}$

$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$ so $m_1 = -1.45$

$m_2 = 0.87.$

update b

$$b^1 = b^0 - \alpha \cdot \frac{dj}{db}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

→ **New error:**

$$\hat{y}^1 = X \cdot m^1 + b^1$$

$$X \cdot m^1 = \begin{bmatrix} 1(-1.45) + 3(0.87) \\ 4(-1.45) + 10(0.87) \end{bmatrix} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

$$\hat{y}^1 = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$e^1 = \hat{y} - y = \begin{bmatrix} 2.15 \\ 3.75 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$\frac{\hat{y} - y}{n} = \frac{1}{2} \begin{matrix} -2.85 \\ -2.21 \end{matrix}$$

$$J = \frac{1}{2}(2.85^2 + 2.21^2 = \frac{1}{2}(8.1225 + 4.8841) =$$

$$\boxed{6.5033}$$

After iteration 1

$m = -1.45, 0.85$

$b = 0.99, 0.89$

$y^1 = 2.15, 3.79$

$e = -2.85, -2.21$

$J = 6.5033$

2nd iteration

Given $\alpha = 0.01$

$$m^{(1)} = [-1.225 \quad 1.435]$$

$$b^{(1)} = \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

predicted values $\hat{y} = X(m^{(1)})^T + b^{(1)}$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.225 \\ 1.435 \end{bmatrix} + \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix}$$

$$= \begin{bmatrix} -1.225 + 4.305 \\ -4.9 + 14.35 \end{bmatrix} + \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix}$$

$$= \begin{bmatrix} 3.08 \\ 9.45 \end{bmatrix} \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 4.075 \\ 10.395 \end{bmatrix}$$

error factor $= e = \hat{y} - y$

$$= \begin{bmatrix} 4.075 \\ 10.395 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e = \begin{bmatrix} -0.925 \\ 4.395 \end{bmatrix}$$

Cost function $J(m, b) = \frac{1}{2} \sum_{i=1}^{2} (\hat{y}_i - y_i)^2$

$$J = \frac{1}{2}[(-0.925)^2 + (4.395)^2]$$

$$= \frac{1}{2}(0.855625 + 19.316025)$$

$$= \frac{1}{2}(20.17165)$$

$$J = 10.085825$$

Gradient with respect to m

$$\nabla_b J = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -0.925 \\ 4.395 \end{bmatrix}$$

$$= \begin{bmatrix} -0.925 + 17.58 \\ -2.775 + 43.95 \end{bmatrix}$$

$$= \begin{bmatrix} 16.655 \\ 41.175 \end{bmatrix}$$

Gradient with respect to b.

$$\nabla_b J = e$$

$$= \begin{bmatrix} -0.925 \\ 4.395 \end{bmatrix}$$

Scanned with CamScanner

update m

$$m^{(2)} = m^{(1)} - \alpha(\bar{V}_m)^T$$

$$= [-1.225 \quad 1.435] - 0.01 [16.665 \quad 41$$

$$= [-12.225 \quad 1.435] - [0.16655 \quad 0.41775]$$

$$m^{(2)} = [-1.39155 \quad 1.02325]$$

update b.

$$b^{(2)} = b^{(1)} - \alpha \bar{V}_b J$$

$$= \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix} - 0.01 \begin{bmatrix} -0.925 \\ 4.395 \end{bmatrix}$$

$$= \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix} - \begin{bmatrix} 0.00925 \\ 0.04395 \end{bmatrix}$$

$$= b^{(2)} = \begin{bmatrix} 1.00425 \\ 0.90105 \end{bmatrix}$$

final result →

$$m^{(2)} = [-1.39155 \quad 1.02325]$$

$$b^{(2)} = \begin{bmatrix} 1.00425 \\ 0.90105 \end{bmatrix}$$

$$J = 10.085825$$

Scanned with CamScanner

$$\hat{y}_2 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.33 \\ 1.17 \end{bmatrix} + \begin{bmatrix} 1.01 \\ 0.92 \end{bmatrix}$$

$$\hat{y}_2 = \begin{bmatrix} 3.21 \\ 7.34 \end{bmatrix}$$

$$MSE_2 = \frac{(\hat{y}_1 - y)^2 + (\hat{y}_2 - y)^2}{2}$$

$$MSE_2 = \frac{(3.21 - 5)^2 + (7.34 - 6)^2}{2}$$

$$= 2.49$$

Iteration 3

$$m_2 = \begin{bmatrix} -1.33 \\ 1.17 \end{bmatrix} \quad b_2 = \begin{bmatrix} 1.01 \\ 0.92 \end{bmatrix}$$

$$e_3 = \hat{y}_2 - y = \begin{bmatrix} 1.01 \\ 0.92 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e_3 = \begin{bmatrix} -3.99 \\ -5.07 \end{bmatrix}$$

$$e_3 = \hat{y}_2 - y = \begin{bmatrix} 3.21 \\ 7.34 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e_3 = \begin{bmatrix} -1.79 \\ 1.34 \end{bmatrix}$$

weight gradients

$$\frac{dE}{dm} = \frac{2}{n} X^T \cdot e_3$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.79 \\ 1.34 \end{bmatrix}$$

$$\frac{dE}{dm} = \begin{bmatrix} 3.59 \\ 8.09 \end{bmatrix}$$

$$\frac{d\bar{e}}{db} = \frac{2}{n} e_3$$

$$\frac{d\bar{e}}{db} = e_3 = \begin{bmatrix} 1.79 \\ 1.34 \end{bmatrix}$$

$$\frac{d\bar{e}}{db} = \begin{bmatrix} 1.79 \\ 1.34 \end{bmatrix}$$

$$m_{new3} = m_{old_2} - \alpha \frac{d\bar{e}}{dm}$$

$$m_{new3} = \begin{bmatrix} -1.33 \\ 1.17 \end{bmatrix} - 0.01 \begin{bmatrix} 3.59 \\ -8.09 \end{bmatrix}$$

$$m_{new3} = \begin{bmatrix} -1.36 \\ 1.09 \end{bmatrix}$$

$$b_{new3} = b_{old_2} - \alpha \frac{d\bar{e}}{db}$$

$$b_{new3} = \begin{bmatrix} 1.01 \\ 0.92 \end{bmatrix} - 0.01 \begin{bmatrix} 1.79 \\ 1.34 \end{bmatrix}$$

$$b_{new3} = \begin{bmatrix} 1.03 \\ 0.89 \end{bmatrix}$$

$$\hat{y}_3 = X_{m_3} + b_3$$

$$\hat{y}_3 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.36 \\ 1.09 \end{bmatrix} + \begin{bmatrix} 1.03 \\ 0.89 \end{bmatrix}$$

$$\hat{y}_3 = \begin{bmatrix} 2.95 \\ 6.37 \end{bmatrix}$$

$$MSE_3 = \frac{(\hat{y}_3 - y)^2 + (\hat{y}_3 - y)^2}{2}$$

$$= \frac{(2.95 - 5)^2 + (6.37 - 6)^2}{2}$$

$$= 2.16$$

Final answers

	m_1	m_2	b_1	b_2
0				
1				
2				
3				

Final answers (bis a vector)

	m	b	MSE
0	$\begin{bmatrix} -1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	61
1	$\begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$	$\begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$	6.5
2	$\begin{bmatrix} -1.33 \\ 1.17 \end{bmatrix}$	$\begin{bmatrix} 1.01 \\ 0.91 \end{bmatrix}$	2.49
3	$\begin{bmatrix} -1.36 \\ 1.09 \end{bmatrix}$	$\begin{bmatrix} 1.03 \\ 0.89 \end{bmatrix}$	2.16